

Q73 - Constraint-Based Pattern Matching (NP-Complete Problem)

Aim: To perform pattern matching under positional constraints using backtracking.

Constraints: Alignment rules and matching constraints.

Approach: Use backtracking to explore feasible alignments and prune invalid matches.

Pruning: Stop exploring when mismatch or constraint violation occurs.

Complexity: Exponential in worst case.

NP-Complete: Multiple constrained alignments make the problem NP-Complete.

Conclusion: Backtracking with pruning improves feasibility and reduces search space.

Output Screenshot: Insert screenshot here.